

What Does a Glucose Foundation Model Understand About Me?

A Single-Person Test of Frozen OpenCGM-StateEvent Embeddings

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Abstract

A frozen glucose encoder did not establish added forecasting value beyond eight glucose/time summaries in this retrospective self-study. The experiment evaluates one public OpenCGM-StateEvent checkpoint, an independent community reconstruction of the GlucoFM method, on the author’s continuous glucose monitoring history. Fifteen CareLink exports contain 338,084 distinct local glucose timestamps across May 2023–September 2026. An audit identified 24 conflicting timestamps on two clock-change days; a conservative rule excluded windows touching any recorded clock-change day. The resulting 11,044 eligible windows support a temporally separated comparison of persistence, glucose-summary Ridge, and the same Ridge predictor augmented with 128 frozen encoder features. Preprocessing, splits, tuning and the primary outcome were recorded before scores were inspected. On 654 later-test windows across 12 weeks, 60-minute mean absolute error was 42.59 mg/dL for persistence, 37.66 for summary Ridge and 37.36 for augmented Ridge. The paired augmented-minus-summary difference was -0.30 mg/dL (95% week-bootstrap interval -0.87 to $+0.34$). All eight horizon-by-period intervals included zero; several point estimates favored the simpler model. The result is limited to one individual, one community checkpoint and one forecasting protocol; it is neither an evaluation of Google’s weights nor evidence of equivalence or physiological understanding.

1. Introduction

At the primary 60-minute horizon, a frozen 128-dimensional glucose embedding reduced error by only 0.30 mg/dL beyond eight glucose and time-of-day summaries, with uncertainty spanning both improvement and deterioration. That narrow result answers the practical question behind this study: did the representation earn its place in a predictor of this person’s glucose?

The author reports living with type 1 diabetes for 17 years and managing his own closed-loop therapy. GlucoFM [1] provided the motivation to ask whether a model learned from other people’s glucose recordings could add useful information about his own history. The paper reports a dual-stream representation of slower glucose state and faster events, followed by frozen-feature downstream evaluations. Its reported results are not reproduced here.

A bounded source search did not locate a verified official Google checkpoint. The present experiment therefore uses the explicitly identified OpenCGM-StateEvent community release [2, 3]. Its model card describes an independent reconstruction trained on public cohorts. It is a distinct empirical object: evaluating it does not test Google’s checkpoint or establish whether the original paper’s results reproduce.

The operational question is deliberately smaller than the title: **do frozen encoder features improve a linear predictor that already receives summaries of the same observed glucose history?** All arms share issue times, history, outcomes and temporal splits. The comparison tests incremental forecasting utility within a fixed model family. It does not test the causal meaning of the representation or the author’s response to model advice.

The contribution is a measured negative result with an auditable chain from export semantics to paired forecast errors. Two details matter to that chain: conflicting wall-clock timestamps cannot be resolved by silently keeping the first row, and midnight insulin aggregates cannot stand in for timestamped deliveries. These observations determined the eligible data before any encoder comparison score was examined.

Evidence convention. Numerical findings in this manuscript are measured outputs of the commands in Appendix B. Claims about the original model and community pretraining are attributed to their sources. Missing ordering evidence and limits of inference are stated explicitly. The evaluation specification was frozen locally in `TANDEM.md` before scoring; it was not an externally registered protocol.

2. Personal archive and measurement boundaries

2.1. Archive coverage and clock ambiguity

Fifteen non-overlapping CareLink CSV exports contain 402,385 dated event rows and 338,084 distinct local glucose timestamps from 9 May 2023 to 20 September 2026. Glucose readings occur on 1,227 days. The inventory distinguishes coverage inspection from forecasting: inspecting an export does not make a later score an untouched prospective result.

There are 24 timestamps with conflicting glucose values on two days containing recorded pump-clock changes. Across the archive, 17 clock-change rows populate only `Index`, `Date`, `Time` and `New Device Time`. None of the conflicting glucose pairs brackets a clock-change row by export index. Neither those fields nor export order establish a verified monotone acquisition timeline or UTC offset.

The implemented rule rejects any window whose full support, from 24 hours before issue to 122.5 minutes after issue, touches a conflict day or any calendar day between a clock-change row’s old and new dates, inclusive. This excludes 17 calendar days and 399 candidate windows. It is intentionally broader than deleting the 24 colliding timestamps: histories crossing a clock adjustment can have uncertain elapsed-time meaning even when their timestamps are unique. Original records are preserved. All 15 CSV hashes still match the pre-experiment inventory.

2.2. Insulin semantics and scope

All 1,200 `CLOSED_LOOP_AUTO_INSULIN` rows in these exports occur at midnight. They are treated as daily aggregates, not timestamped insulin deliveries. The present comparison uses only glucose and local time of day; neither those aggregates nor carbohydrate or bolus records enter its predictors.

An earlier project experiment evaluated recorded carbohydrates and prior boluses at meal-related issue times. It used a different archive, population and tuning design. Its results and corrected feature errors remain in the historical record, but its scores are not pooled with this comparison. The current fixed-time sampling also avoids excluding a forecast because another meal will occur after it. Requiring complete future glucose observation still introduces selection, as discussed in Section 5.

3. Evaluation design

3.1. History, outcomes and eligibility

Forecasts are issued every two hours on the pump-local wall clock. Let t be an issue time and $s = t - 24$ h its history start. Each model receives the same oldest-first 288-position history. Observed glucose is converted from mmol/L to mg/dL using the existing conversion factor 18.0182. Source timestamps in $[s, t)$ are assigned to positions by

$$j = \left\lfloor \frac{\tau - s}{5 \text{ min}} + \frac{1}{2} \right\rfloor, \quad j \in \{0, \dots, 287\}.$$

Readings assigned to the same position are averaged; indices outside this range are discarded. This follows the pinned source implementation [3]. Missing positions have value zero and mask zero. Observed positions have mask one. There is no interpolation or external per-window normalization. The clock exclusions are applied before binning, so conflicting raw timestamps are never averaged into a model input.

Eligibility requires at least 80% observed bins over the full history and its final three hours, an observed final bin, and no raw history gap longer than 60 minutes, including the history edges. The minimum full-history count among retained windows was 270 of 288 observed bins; the other rules can be stricter than the nominal coverage threshold.

Targets are means of observed readings in half-open bins $[t + h - 2.5 \text{ min}, t + h + 2.5 \text{ min})$. All 24 five-minute future bins must be observed. The evaluated horizons are 30, 60, 90 and 120 minutes, with identical samples at every horizon. Targets therefore have a stated temporal tolerance; they are not asserted to be exact-time measurements. The last history bin is centered five minutes before issue, so persistence also uses that bin rather than a new reading at issue time.

Table 1. Measured eligibility flow. Exclusions are sequential and mutually exclusive in the listed order.

Stage	Removed	Remaining
Fixed two-hour candidate issue times	—	14,754
Incomplete archive-boundary support	14	14,740
Clock-ambiguity support overlap	399	14,341
History gap longer than 60 minutes	2,939	11,402
History coverage or final-bin failure	103	11,299
Missing future target bins	255	11,044

3.2. Checkpoint identity and feature extraction

The evaluated artifact is `sfourdrinier/opencgm-stateevent`, Hugging Face revision `6c53e572c2dafcb0b030f2e1aa4f3a2a8d7f7ca3`. Its ONNX encoder is byte-identical to the encoder stored at GitHub source revision `bcf8ac2dfca00f8fb3c7728fb2a3a2bd2803bebf`. Appendix A records the full encoder digest and runtime contract.

The maintainer identifies this artifact as seed 17, epoch 40, trained with raw glucose statistics and `zero_empty_patches=false` [2]. The card’s separate five-seed, 120-epoch evaluation is not evidence about the exact file used here. Its public-cohort pretraining description remains maintainer-reported; this study verifies file identity and local behavior, not the training history.

Inputs are float32 glucose and mask arrays of shape $B \times 288$, plus an int64 local circadian start index of shape B . The released graph performs observed-only instance normalization, state/event processing and contextual encoding. Raw-mg/dL patch statistics also enter the encoder, so externally standardizing glucose would change its documented contract. The returned $B \times 128$ embedding is the unweighted mean of 24 contextual tokens from the frozen online branch. We use the released ONNX graph directly; rebuilding with current source defaults would change its empty-patch behavior.

Synthetic graph and inference checks preceded private inference. Constant, ramp, sparse, empty and partially masked histories produced finite embeddings. Repeated calls were deterministic; batch-size and finite masked-fill checks agreed. All private inference then ran locally through the CPU provider. No encoder parameter or bundled probe head was fitted to the author’s records.

3.3. Matched predictors and temporal fitting

Persistence predicts the final observed history-bin value at every horizon. **Summary Ridge** uses eight features: that value; observed-history mean, sample standard deviation, minimum and maximum; a least-squares slope over the last six bins in mg/dL per minute; and sine/cosine of issue time. **Augmented Ridge** uses those same eight features plus all 128 frozen embedding dimensions.

Each Ridge model minimizes

$$\sum_{i \in \mathcal{T}} (y_{i,h} - b_h - z_i^\top \beta_h)^2 + \alpha_h \|\beta_h\|_2^2,$$

where z_i contains columns standardized with training-set means and population standard deviations. The intercept is unpenalized. There is no PCA, target normalization or prediction clipping. Each arm and horizon selects its own regularization value from the same 19 log-spaced candidates, 10^{-3} through 10^6 , by minimum event-weighted validation MAE; ties select the smaller value.

Table 2. Measured temporal populations. Every boundary condition applies to complete history or target support, not merely the issue date.

Population	Support condition	Windows
Development training	Target support ends before 1 Jan 2025	5,334
Validation	History starts on/after 1 Jan 2025; target support ends before 1 Jan 2026	3,357
Final training	All target support ends before 1 Jan 2026	8,704
Exploratory test	History starts on/after 1 Jan 2026; target support ends before 1 Jul 2026	1,662
Later retrospective test	History starts on/after 1 Jul 2026; targets available in the archive	654

After tuning, scaling and Ridge coefficients are refitted on final training only and frozen for both test periods. Thirteen windows purged between development training and validation rejoin final training; 24 eligible windows at test boundaries remain unused. Complete-support purging prevents shared readings across the fitted-training/test boundary. Overlap within each temporal population remains. Earlier forecasting work examined March–June 2026, so the whole January–June test is labeled exploratory. July–September had coverage and export inspection but no recorded prior forecast comparison. It is a later retrospective test, not a pristine prospective holdout. The specification was recorded before comparison scores were inspected, and no score-driven change to the protocol followed.

3.4. Primary estimand and uncertainty

The primary outcome is the later-test difference in 60-minute MAE between augmented and summary Ridge:

$$\Delta_{60} = \frac{1}{n} \sum_{i=1}^n \left(|\hat{y}_{i,60}^{\text{aug}} - y_{i,60}| - |\hat{y}_{i,60}^{\text{sum}} - y_{i,60}| \right).$$

Negative values favor embeddings. Each test slice uses 2,000 paired bootstrap resamples of ISO

calendar weeks, seed 42. A sampled week contributes all of its events, and estimates remain event-weighted rather than averaging week means. The same draws serve every arm and horizon within the slice. Reported intervals are the 2.5th and 97.5th percentiles.

These intervals condition on the fitted predictors. They do not include uncertainty from selecting or retraining the models, and week clustering does not guarantee that adjacent weeks are independent. Secondary horizons are descriptive; there is no multiplicity-adjusted superiority or equivalence claim.

4. Results

4.1. Later retrospective test

The primary test contains 654 windows in 12 weeks. At 60 minutes, summary Ridge achieved 37.66 mg/dL MAE and augmented Ridge 37.36 mg/dL, compared with 42.59 mg/dL for persistence. The paired augmented-minus-summary difference was **−0.30 mg/dL [−0.87, +0.34]**. This interval includes both a modest improvement and a modest deterioration.

Embeddings increased 30-minute error by 0.08 mg/dL and decreased the 90- and 120-minute errors by 0.36 and 0.11 mg/dL. Every paired interval includes zero (Table 3; Figure 1). Both Ridge arms improved on persistence at every horizon in this slice; the paired summary-minus-persistence difference at 60 minutes was **−4.93 mg/dL [−6.30, −3.47]**. The summaries account for a clear gain over that reference, while the added representation does not establish a further gain.

Table 3. Measured forecast errors and paired effects. MAE and intervals are in mg/dL. P: persistence; S: summary Ridge; A: augmented Ridge. Negative A−S favors embeddings. The later 60-minute comparison is primary.

Slice	Horizon	P MAE	S MAE	A MAE	A−S [95% interval]
Later	30 min	26.76	22.50	22.58	+0.08 [−0.34, +0.51]
Later	60 min	42.59	37.66	37.36	−0.30 [−0.87, +0.34]
Later	90 min	54.71	44.92	44.56	−0.36 [−0.77, +0.13]
Later	120 min	60.83	47.51	47.41	−0.11 [−0.65, +0.44]
Exploratory	30 min	26.51	23.06	23.05	−0.02 [−0.26, +0.21]
Exploratory	60 min	42.14	37.17	37.33	+0.17 [−0.20, +0.51]
Exploratory	90 min	52.12	44.87	45.02	+0.15 [−0.36, +0.64]
Exploratory	120 min	57.21	47.57	47.79	+0.22 [−0.36, +0.81]

4.2. Exploratory test and tuning outcomes

The exploratory slice contains 1,662 windows in 27 weeks. Augmented Ridge increased MAE at 60, 90 and 120 minutes by 0.17, 0.15 and 0.22 mg/dL, respectively. At 30 minutes it reduced MAE by 0.02 mg/dL. All four intervals include zero. These losses receive the same reporting and uncertainty calculation as the later-slice improvements.

Validation selected the smallest allowed alpha, 0.001, for summary Ridge at all horizons. Augmented Ridge selected 31.6228 at 30 minutes and 100 at the other horizons. The predefined grid was not expanded after examining results. Consequently, the comparison describes performance under this tuning budget, not an exhaustive search over downstream models.

4.3. Verification of the measurement chain

Seven synthetic contract tests passed: parity with upstream binning and exclusion of future input; half-open target boundaries; clock-day support exclusion and conflict preservation; temporal support

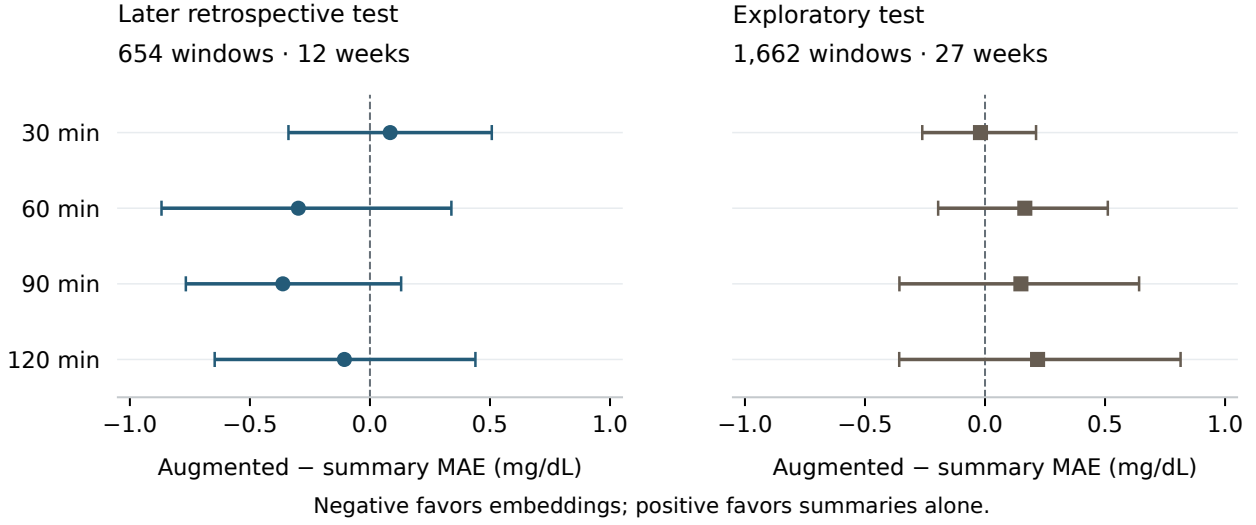


Figure 1: Paired augmented-minus-summary MAE at all four horizons. Points are measured differences; lines are 95% percentile intervals from 2,000 paired week-bootstrap draws. Both panels use the same scale. All eight intervals include zero; the later 60-minute comparison is primary.

purging; training-only scaling/fitting with test-target perturbation; paired event-weighted bootstrap agreement; and private-output path enforcement.

A separate saved-artifact verifier checked every retained window against clock exclusions. It reproduced 64 stratified private histories using the pinned upstream grid implementation and independently matched 256 horizon targets to observed-bin means. Saved prediction identities, targets and persistence values matched the input windows. All reported MAEs, paired differences and intervals matched an explicit-row implementation of week resampling. These are implementation and arithmetic checks; they do not validate the checkpoint’s claimed pretraining history or resolve observational limitations.

5. Interpretation and limitations

Incremental utility, not absence of information. The experiment found no clear added forecasting value from this frozen representation under the chosen Ridge readout. It does not show that the embedding contains no useful information. A linear predictor over a mean-pooled daily representation tests a particular route from representation to forecast. The augmented arm also has more input dimensions, so the result reflects the representation and its fitted readout together.

A conditional negative result, not equivalence. Small point estimates and intervals spanning zero do not prove equal performance. Only 12 weeks contribute to the later test. The reported uncertainty omits training variation, alternative split choices and possible dependence across weeks. No clinically meaningful equivalence margin was specified.

Selection and availability. Complete future target observation is known only retrospectively, and history eligibility favors well-recorded periods. The resulting errors do not estimate continuous deployment performance over missing or ambiguous intervals. Export timestamps also do not establish when each reading would have reached a live prediction system. The last-bin convention gives all arms the same available input here, but it is not a claim about optimal real-time timing.

One person and a changing treatment context. Thousands of windows from one person do not constitute thousands of independent participants. The glucose trajectory was recorded under the author’s ongoing treatment and behavior. This study neither randomizes that context nor evaluates

model-guided treatment changes. It cannot identify causes of glucose changes, infer insulin sensitivity from prediction accuracy, or establish transfer to other people.

One community artifact and a different task. The released weights are a single checkpoint with maintainer-reported provenance. Hash matching establishes which bytes were evaluated. It does not make the artifact an official GlucoFM release, independently verify its training corpus, or turn this fixed-time forecasting task into a reproduction of the original phenotype or meal-response evaluations [1, 2].

6. Conclusion

On the author’s eligible later glucose windows, eight glucose/time summaries supported a stronger predictor than persistence, while adding the pinned OpenCGM-StateEvent embedding produced only small, inconsistent changes. The primary 60-minute difference was -0.30 mg/dL, with a paired interval spanning zero. The defensible conclusion is specific: **this frozen community checkpoint did not establish added predictive value beyond these summaries under this evaluation**. The broader question of physiological understanding remains unanswered by forecast accuracy alone.

Data, code and assistance

The author is the sole participant and data source. Raw CareLink exports, window-level features, embeddings and predictions remain private. All inference was local. The accompanying project files contain the analysis code, pinned source and artifact references, synthetic tests, and non-reconstructive aggregate outputs. Complete numerical replication requires the private inputs; the manuscript and its figure can be rebuilt from aggregates alone. A permanent public code-archive identifier has not been assigned at this checkpoint.

Code, analysis checks and manuscript preparation used AI assistance. Numerical claims were checked against executable outputs and cited source records; agreement between assistants was not treated as validation. The earlier baseline article and its corrections remain in the project history.

References

1. Zechen Li et al. *GlucoFM: A Dual-Stream Foundation Model for Continuous Glucose Monitoring*. arXiv:2605.30865, version 2, 25 August 2026. <https://arxiv.org/abs/2605.30865v2>.
2. Stephane Fourdrinier. *OpenCGM-StateEvent: model card and released encoder*. 2026. Hugging Face repository `sfourdrinier/opencgm-stateevent`, revision `6c53e572c2dafcb0b030f2e1aa4f3a2a8d7f7ca3`. [Pinned model card](#).
3. Stephane Fourdrinier. *OpenCGM: source and ONNX export implementation*. 2026. GitHub repository `sfourdrinier/opencgm`, revision `bcf8ac2dfca00f8fb3c7728fb2a3a2bd2803bebf`. [Pinned source](#).

Appendix A. Exact encoder and protocol identity

Encoder SHA-256:

b1349deffd15ab62a5a98d7c7c4a7e143bcf1dee7ad745b1e05533ff54f34768

Frozen protocol-section SHA-256:

3cc2fd06755d644bdc1a44efdb83d17c62922472b4a06ad8183569a2f69853bb

The encoder file is 1,991,782 bytes, opset 17. Its inputs are `values` and `mask`, float32 arrays [B,288], and `circadian_start`, int64 [B]. The start index is the integer part of local minutes since midnight divided by five. The output is `embedding`, float32 [B,128]. Missing values are finite zeros with mask

zero; NaN fill is not part of the tested contract.

For observed mask m_j , values x_j and $n = \sum_j m_j$, the source’s internal normalization uses

$$\mu = \frac{\sum_j m_j x_j}{n + \epsilon}, \quad v = \frac{\sum_j m_j (x_j - \mu)^2}{n + \epsilon}, \quad \tilde{x}_j = \frac{m_j (x_j - \mu)}{\max(\sqrt{v + \epsilon}, 10^{-4})}, \quad \epsilon = 10^{-6}.$$

An all-unobserved window returns zeros at this normalization step. The full encoder may still emit a nonzero embedding. The source paths establishing the contract are `data/grid.py`, `data/timestamps.py`, `model/causal_gaussian.py`, `model/reference.py` and `model/encoder.py` under `src/opencgm_stateevent/`, plus `scripts/export_encoder_onnx.py`. Their individual hashes and the complete artifact metadata are recorded in `assets/article/opencgm-provenance.json`.

The model card labels the encoder CC-BY-NC-4.0 and the source Apache-2.0. It reports pretraining on Stanford, Shanghai T2DM, Big Ideas and Colas, while recording unresolved source licensing for Stanford [2, 3]. The separate bundled probe heads are not used in this study.

Appendix B. Reproduction and evidence map

Commands below run from the project root using the recorded Python 3.14.3 environment. Runtime dependencies are pinned in `carelink-local/requirements-opencgm.txt`: NumPy 2.5.3, pandas 3.0.6, ONNX 1.23.0 and ONNX Runtime 1.30.0. The measured runtime is macOS arm64 with the CPU execution provider.

```
PY=carelink-local/.venv/bin/python
$PY carelink-local/archive_inventory.py private-data/carelink/*.csv \
  --output assets/article/archive-inventory.json
$PY carelink-local/clock_audit.py
$PY carelink-local/prepare_opencgm.py
$PY carelink-local/opencgm_comparison.py smoke
$PY -m unittest discover -s carelink-local \
  -p 'test_opencgm_comparison.py' -v
$PY carelink-local/opencgm_comparison.py build
$PY carelink-local/opencgm_comparison.py score
$PY carelink-local/verify_opencgm_run.py
```

The inventory command reproduces the archive audit; the comparison commands correspond to the executed stages. The frozen protocol remains in `TANDEM.md`. Aggregates under `assets/article/` provide the evidence chain: `archive-inventory.json` for source coverage and midnight aggregates; `clock-audit.json` for collisions and exclusions; `opencgm-provenance.json` for identity; `opencgm-smoke.json` for synthetic inference; `opencgm-build.json` for eligibility and splits; `opencgm-results.json` for every arm, horizon, interval and chosen alpha; and `opencgm-checks.json` for independent saved-artifact verification.

The results JSON also contains absolute-MAE intervals and paired comparisons against persistence for both periods. All published tables and the figure are checked against that aggregate. Generating the preprint does not rerun model selection, alter the frozen protocol, or read private windows.